

CarbonX — Location-Based Project Evidence

Wayanad Green Corridor Demo — Synthetic Dataset

DEMO / SYNTHETIC DATA ONLY. This dataset is for testing evidence ingestion, heterogeneous-data reconciliation and Truth Score logic. It is not a real carbon-credit project or certification.

1. Project Identification

Field	Value
Project name	Wayanad Green Corridor Demo
Project type	Afforestation
Country / state / district	India / Kerala / Wayanad
Reference centroid	11.688714 N, 76.132857 E
Geometry	Polygon
GIS observed area	64.63 ha
Claimed area	66.56 ha
Claimed carbon	5,457.92 tCO ₂ e
Project start	2022-07-01
Monitoring date	2026-06-22

2. Boundary Evidence

The companion GeoJSON contains a synthetic polygon in Wayanad, Kerala. Approximate GIS area is 64.63 ha. The document claim is 66.56 ha, creating a small boundary discrepancy for reconciliation testing.

3. Carbon Evidence

Component	Value	Role
Headline claim	5,457.92 tCO ₂ e	Project claim
Gross biomass benefit	5,443.81 tCO ₂ e	Synthetic calculation
Baseline adjustment	-410 tCO ₂ e	Deduction
Project emissions	-190 tCO ₂ e	Deduction
Leakage deduction	-150 tCO ₂ e	Deduction
Independent derived value	4,693.81 tCO₂e	Reconciled evidence
Discrepancy	764.11 tCO ₂ e	REVIEW trigger

The independent carbon value is about 14.0% below the claim. CarbonX should flag the mismatch and explain the evidence behind it.

4. Field Monitoring Evidence

Plot	Trees	Height m	Biomass kg	Date
W001	143	5.1	2,610	2026-06-18
W002	136	4.9	2,480	2026-06-18
W003	151	5.3	2,770	2026-06-19
W004	129	4.7	2,290	2026-06-19
W005	147	5.0	2,560	2026-06-20

These values are synthetic field evidence intended to be cross-checked against a future CSV/XLSX upload.

5. Remote-Sensing Evidence

Metric	Observation	Meaning
Mean NDVI	0.71	Strong vegetation signal
Median NDVI	0.73	Central vegetation condition
NDVI std dev	0.09	Moderate spatial variation
Valid pixels	22,740	Synthetic coverage
Cloud-free fraction	94%	Synthetic quality indicator
Observation date	2026-06-22	Monitoring observation

NDVI is supporting vegetation evidence, not a direct carbon measurement. The scoring engine must not equate NDVI directly to tCO₂e.

6. Provenance IDs

Evidence ID	Field	Value	Source
GEO-WAY-001	GIS_AREA_HA	64.63	Companion GeoJSON
PDD-WAY-001	CLAIMED_AREA_HA	66.56	Synthetic PDF
PDD-WAY-002	CLAIMED_TCO ₂ E	5457.92	Synthetic PDF
PDD-WAY-003	DERIVED_TCO ₂ E	4693.81	Synthetic calculation
FIELD-WAY-001	FIELD_PLOTS	5	Synthetic field evidence
SAT-WAY-001	NDVI_MEAN	0.71	Synthetic remote sensing

7. Expected CarbonX Outcome

Expected: geometry is valid and broadly consistent with the claimed area; field and vegetation evidence support an afforestation scenario; however, the carbon claim (5,457.92 tCO₂e) exceeds the independent derived value (4,693.81 tCO₂e). The system should reduce the truth score, produce an explanation, and recommend REVIEW.

8. Limitations

All project values are synthetic. The polygon is a generated test geometry and is not a real project boundary. This document must not be used as evidence of actual carbon sequestration or certification.